

Products – Control (PID Tuning)

Menu: Products > Control

Configures the PID controller that adjusts the motor or valve to maintain the target rate.

Primary Settings

Setting	Description
1. Gain	Rate adjustment factor — how strongly the controller responds to rate errors. The effect is linear: doubling Gain doubles the correction. Errors are measured as a percentage of the target rate, so the same Gain setting behaves consistently at low and high rates.
2. Integral	Uses accumulated error to reach target rate faster
3. Max Power	Maximum PWM power delivered to motor or valve
4. Min Power	Minimum PWM power delivered to motor or valve

Adjustment Process

1. **Initial Setup** — Set minimum power to the minimum needed to move the motor/valve. Set maximum power to 100. Set Integral to 0.
2. **Gain Adjustment** — Slowly increase gain until the system overshoots the target, then reduce slightly to stabilize.
3. **Integral** — If needed, increase integral to reach the target rate more quickly.

Optional / Advanced Settings

Setting	Description
1. Deadband	Error % below which no adjustment is made
2. BrakePoint	Error % where adjustment switches to slow rate
3. Slow Adjustment	% of full adjustment to use at slow rate
4. PID loop time	Adjustment loop interval in milliseconds
5. Slew Rate	Maximum total PWM change per PID loop
6. Max Integral change	Maximum Integral PWM change per PID loop
7. Min Start % (timed combo)	% of target rate for a faster start from 0
8. Adjust time (timed combo)	Milliseconds the valve adjusts
9. Pause time (timed combo)	Milliseconds the valve pauses adjustment
10. Min flow sensor Hz	Minimum valid flow sensor pulse frequency
11. Max flow sensor Hz	Maximum valid flow sensor pulse frequency
12. Flow sample time (Smp Tm)	Time window in milliseconds (50–1500) used to calculate the median flow reading. A longer window smooths the reading; a shorter window responds faster. Default 400 ms.

Controls

Control	Action
 Navigate left	Previous product
 Navigate right	Next product
 Load defaults	Restore default PID values
 Open rate graph	Display the rate graph
 Gain +	Increase Gain value
 Gain -	Decrease Gain value
 Integral +	Increase Integral value
 Integral -	Decrease Integral value
 Max Power +	Increase Max Power value
 Max Power -	Decrease Max Power value
 Min Power +	Increase Min Power value
 Min Power -	Decrease Min Power value
 Show advanced settings	Expand the advanced/optional settings panel
 Cancel	Discard changes
 Save	Save settings